

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 June 2026
Name	Nadin Surya S
Project Name	Meeting Room Booking System using ServiceNow App Engine Studio
Maximum Marks	4 Marks

Technical Architecture

The diagram below shows the technical architecture of the Meeting Room Booking System, built entirely on the ServiceNow App Engine Studio platform, along with the components and technologies used (Table-1) and the application characteristics (Table-2).

Meeting Room Booking System - Architecture Diagram

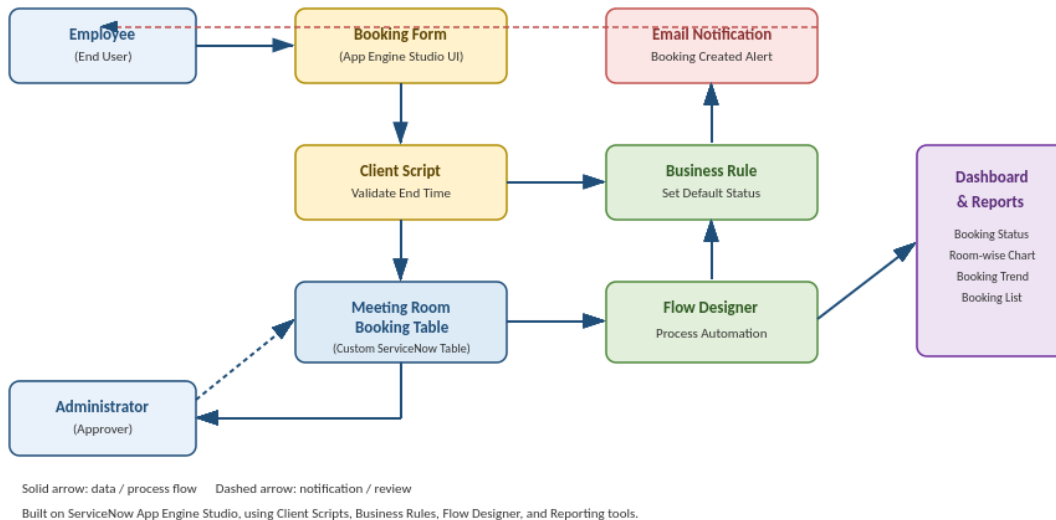


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the employee/admin interacts with the application	App Engine Studio UI Builder, HTML, CSS, JavaScript (Now Platform)
2.	Application Logic-1	Client-side validation of booking time before submission	Client Script (JavaScript — ServiceNow)
3.	Application Logic-2	Auto-assigns default booking status on record creation	Business Rule (Server-side JavaScript)
4.	Application Logic-3	Automates the booking process and triggers notifications	Flow Designer (ServiceNow native automation)
5.	Database	Stores booking records, fields and choice values	ServiceNow Platform Database (custom table)
6.	Cloud Database	Database service hosting the application instance	ServiceNow Now Platform Cloud (multi-instance architecture)
7.	File Storage	Storage for any attachments added to a booking	ServiceNow Attachment Service (sys_attachment)
8.	External API-1	Sends booking confirmation emails to employees	ServiceNow Email Notification (SMTP integration)
9.	External API-2	Authenticates users into the application	ServiceNow SSO / Active Directory Integration
10.	Machine Learning Model	Future scope — predicting room demand/utilization	ServiceNow Predictive Intelligence (future scope)
11.	Infrastructure (Server / Cloud)	Application deployment environment	Local Dev Instance + ServiceNow Production Cloud Instance

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source / Platform Frameworks	UI components are built on the Now Platform's native framework, built on top of AngularJS/Bootstrap-style components	ServiceNow App Engine Studio, Now Experience UI Framework
2.	Security Implementations	Role-based access using Access Control Lists (ACLs); only authenticated users can book, only Admins can approve	ServiceNow ACLs, Role-Based Access Control, Data Encryption (in transit & at rest)
3.	Scalable Architecture	Application is built as a scoped app on a multi-instance cloud platform, allowing it to scale with organizational growth	ServiceNow Multi-Instance Cloud Architecture, Scoped Applications
4.	Availability	ServiceNow platform provides high availability through redundant, geographically distributed data centers	ServiceNow Cloud Infrastructure (99.8% uptime SLA)
5.	Performance	Business Rules and Flows run asynchronously where possible; indexed table queries keep form load and dashboard rendering fast	Asynchronous Business Rules/Flows, Indexed Database Queries, Platform Caching